IES

GenAI for Engineers

One-day Copilot in Excel workshop: SPC, FMEA and DMAIC



Copilot in Excel

SPC • FMEA • DMAIC

ONE DAY | 09:00–17:00

One workbook, three engineering outputs

Reuse evidence across the entire day.



SPC

Understand data and process behaviour

FMEA

Review potential failure and controls

DMAIC

Frame improvement and follow-up

Today’s scope

330 minutes of hands-on Excel work.

Lab01

SPC ·140 minutes

Lab02

FMEA ·100 minutes

Lab03

DMAIC ·90 minutes

Review

Setup30 + final review30 minutes

What Copilot contributes

The engineer verifies the source, formula and interpretation.



Suggest

Tables, formulas and charts

Explain

Patterns and evidence gaps

Review

You check calculations and claims

Prompt a workbook operation

Name the table, output and acceptance test.

Role

Engineering data reviewer

Input

Named Excel table and source revision

Steps

Audit, calculate, chart, check

Expected

Formula-based result with limitations

Constraints make the result reviewable

Specify what must be preserved.

Source

RawTests remains unchanged

Data

Units, time order and missingness

Formula

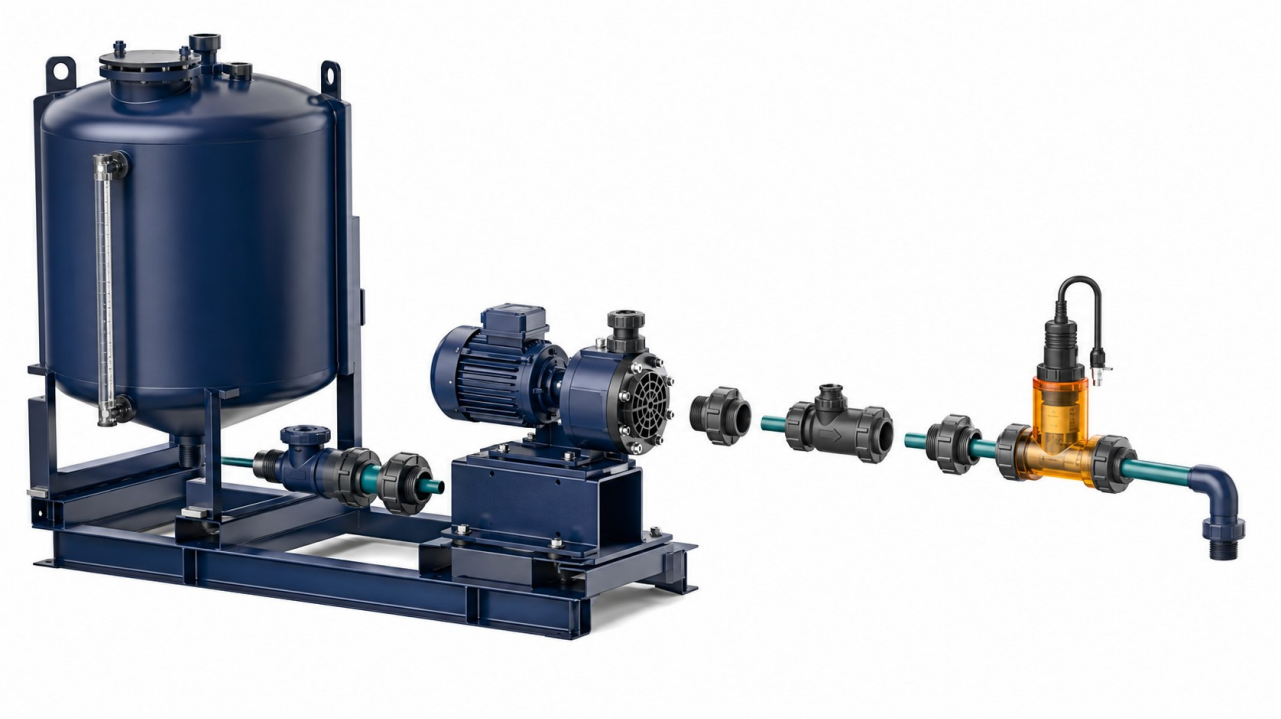
Show rows and parameters

Decision

No invented cause or approval

The CW-401 delivery-test case

One dosing skid; linked measurements and maintenance evidence.



Target

62.0 L/h in a standardised test

Comparison

58.9–65.1 L/h fictional specification

Boundary

Not a chemical compliance limit

Read the evidence before calculating

Separate observation, hypothesis and intervention.

Observed

Delivery falls; differential pressure rises

Hypothesis

Restriction is possible, not proven

Intervention

Strainer change plus verification

Gap

Effects of the two changes not isolated

Prepare Excel for the session

Use approved storage and confirm Copilot access.

Workbook

Save a working copy of Starter.xlsx

Source

Select the relevant named table

Copilot

Use the Excel side pane

Fallback

Record manual formulas honestly

Lab01 · Data quality and SPC

140 minutes | QualityAudit, formulas, chart and interpretation.

01

Inputs

RawTests, ValidTests and Parameters

02

Deliverable

Fixed-baseline individuals chart

03

Check

Every exclusion and formula traceable

SPC1 · Audit the source

Copy from the workbook Prompts sheet into Copilot in Excel.

COPY-READY PROMPT

Using RawTests, audit test_id uniqueness, missing collected_L, duration_s and quality_flag. Do not modify RawTests. Create a QualityAudit table with record ID, issue, inclusion decision and reason. Use one unique record per test, require VALID and 120 seconds, and never discard an excursion solely because it is extreme. Report raw, unique and valid counts.

Input

Select the named source table

Review

Check the changed cells, formulas and claims

SPC2 · Calculate the baseline

Copy from the workbook Prompts sheet into Copilot in Excel.

COPY-READY PROMPT

Use ValidTests in time order. In Analysis create delivery_L_h = collected_L*3600/120. Use only the 60 baseline records to calculate mean and successive moving ranges; do not bridge across phases. Estimate sigma = MRbar/1.128 and individuals limits mean +/-3*sigma. Put constants and specifications in Parameters and show formulas, units and referenced rows.

Input

Select the named source table

Review

Check the changed cells, formulas and claims

SPC3 · Plot and interpret

Copy from the workbook Prompts sheet into Copilot in Excel.

COPY-READY PROMPT

Create an Excel line chart of delivery_L_h by timestamp for baseline, monitoring and pilot, with the fixed baseline mean/UCL/LCL. Flag points outside the limits without refitting them. Report phase counts and affected IDs. Explain the monitoring shift as a signal requiring investigation, not a proven restriction diagnosis. Keep invalid and missing tests visible in QualityAudit.

Input

Select the named source table

Review

Check the changed cells, formulas and claims

SPC4 · Check capability

Copy from the workbook Prompts sheet into Copilot in Excel.

COPY-READY PROMPT

Using fictional test specifications 58.9–65.1 L/h, calculate baseline Cp and Cpk from the stated within-process estimator. State assumptions: stable baseline, adequate measurement, suitable distribution and independence. Do not use a good Cp to claim the later monitoring period is stable. Review potential shift/measurement effects and compare with Worked-Reference.xlsx.

Input

Select the named source table

Review

Check the changed cells, formulas and claims

An export is not an analysis table

Keep a ledger of inclusion decisions.

121

Raw rows

120

Unique test IDs

118

Comparable valid tests

Three issues, three explicit decisions

Do not delete an excursion simply because it is extreme.

01

TEST-068

Missing volume: not zero

02

TEST-083

110s duration: invalid for this protocol

03

TEST-088

Exact duplicate: retain one event

The conversion must preserve the unit

A120-second collection is1/30 of an hour.

Volume

Collected litres

Scale

3600 /120 =30

Delivery

Volume ×30 L/h

Freeze the reference period

Monitoring and pilot data do not set these limits.



Baseline

60 valid tests

Monitoring

28 valid tests

Pilot

30 valid tests

Build the individuals-chart limits

Use successive baseline moving ranges only.

Mean

62.00250 L/h

MRbar

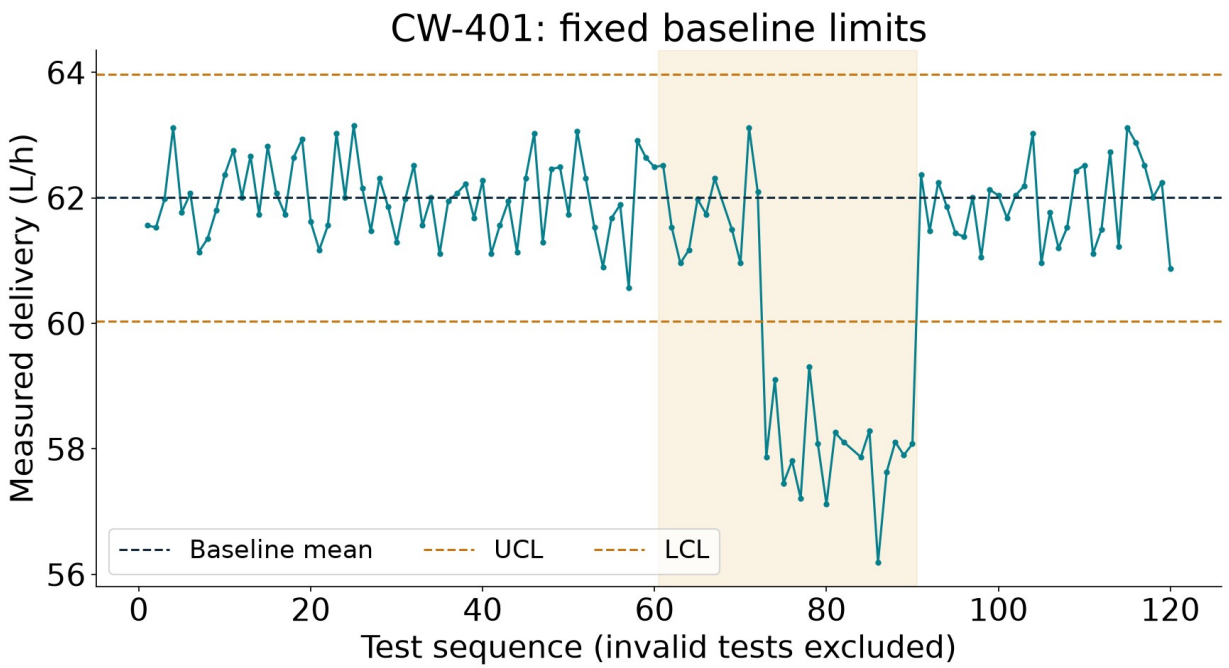
0.73881 L/h

Sigma

MRbar /1.128 =0.65498

The monitoring period shifts

Fixed limits reveal a change that requires investigation.



LCL

60.03757 L/h

UCL

63.96743 L/h

Signal

17 monitoring points below LCL

Specifications and control limits differ

Requirements and observed behaviour answer different questions.

01

Specification

58.9–65.1 L/h for this teaching test

02

Control limits

Estimated from the chosen baseline

03

Interpretation

Neither alone establishes physical cause

Capability is conditional

Do not use baseline capability to dismiss later instability.

Cp

1.57766

Cpk

1.57639

Conditions

Stable behaviour and suitable measurement

A good SPC conclusion has a boundary

A signal is evidence for investigation.

01

Supported

Monitoring delivery is below baseline limits

02

Associated

Differential pressure rose in the same period

03

Unresolved

Measurement adequacy and sole physical cause

Lab02 · FMEA in Excel

100 minutes | Six rows, formula checks, priority chart and review.

01

Inputs

FMEA_Input, ScoringNote, Evidence

02

Deliverable

FMEA_Review and action view

03

Check

Ratings, source status and uncertainty

FMEA1 · Structure the review

Copy from the workbook Prompts sheet into Copilot in Excel.

COPY-READY PROMPT

Use FMEA_Input and Evidence. Create FMEA_Review with function, mode, effect, potential cause, current control, evidence, supplied provisional S/O/D, RPN, action, owner and status. Keep the six mode IDs. Separate actual observations from prospective scenarios. Use ScoringNote for definitions and do not infer occurrence ratings from the short test series.

Input

Select the named source table

Review

Check the changed cells, formulas and claims

FMEA2 · Calculate and prioritise

Copy from the workbook Prompts sheet into Copilot in Excel.

COPY-READY PROMPT

Add an Excel formula RPN=S*O*D for every row. Sort a view by RPN descending while preserving mode IDs and source rows. Add a separate high-severity review flag for S>=9. Create an RPN column chart. Do not treat rank or low RPN as evidence of safety acceptance.

Input

Select the named source table

Review

Check the changed cells, formulas and claims

FMEA3 · Challenge with personas

Copy from the workbook Prompts sheet into Copilot in Excel.

COPY-READY PROMPT

Simulate reliability, operations and safety review lenses. First return one separate evidence-based concern per role. Run two concise challenge/response rounds, citing mode IDs and evidence records. A moderator returns agreed revisions, unresolved disagreement and missing evidence. Do not force consensus or invent approval. Keep residual ratings unassessed until action-effectiveness evidence exists.

Input

Select the named source table

Review

Check the changed cells, formulas and claims

Keep the failure story distinct

One row should remain logically connected.



Function	Mode	Effect	Cause
Meter delivery	Low delivery	Insufficient feed	Restriction hypothesis

Scores are provisional judgements

This small dataset does not estimate a failure probability.

S

Effect severity

O

Occurrence judgement

D

Poorer detection = larger score

Basis

Supplied workshop definitions

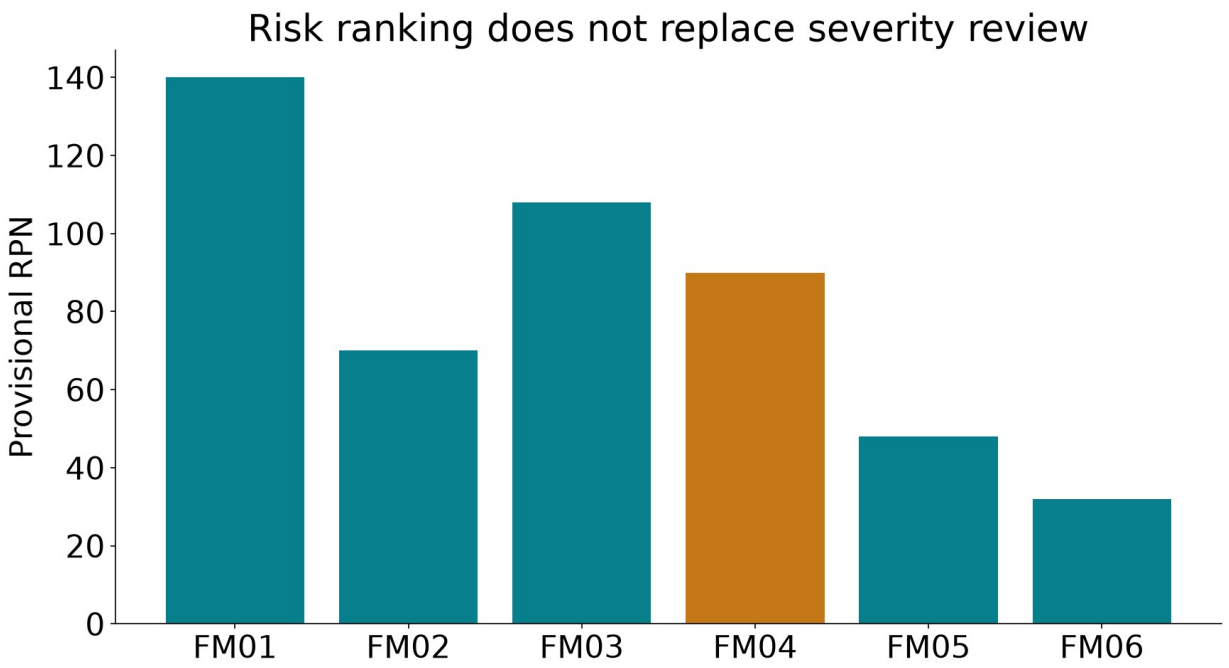
Use formulas, then verify the IDs

FM01 provides a simple arithmetic check.

Severity	Occurrence	Detection	RPN
7	5	4	7 ×5 ×4 =140

Severity review can outrank RPN

FM04 has S = 9 and still needs explicit review.



FM01

Highest RPN 140

FM03

RPN 108

FM04

RPN 90; severity 9

Current controls are not future actions

A proposed improvement has not yet demonstrated effectiveness.

01

Current

Periodic collection test

02

Proposed

Review mechanism and measurement evidence

03

Residual

Unassessed until verified evidence

Use three review lenses

Simulated personas expose questions, not independent approval.

01

Reliability

What supports the mechanism?

02

Operations

What is practical and missing?

03

Safety

What consequence needs review?

Debate without forcing agreement

Two short rounds keep the review focused.



Position

Separate evidence-based concerns

Challenge

One specific objection per lens

Response

Cite mode and source record

Moderator

Retain dissent and evidence gaps

Close an action with evidence

A complete spreadsheet is not a completed control.

Owner

Responsible role identified

Evidence

Effectiveness test specified

Status

Open until reviewed

Scores

No automatic residual reduction

Lab03 · DMAIC in Excel

90 minutes | Rates, Pareto, improvement proposal and control plan.

01

Inputs

WeeklyEvents and DefectCounts

02

Deliverable

DMAIC and ControlPlan sheets

03

Check

Denominators and limitations retained

DMAIC1 · Define and measure

Copy from the workbook Prompts sheet into Copilot in Excel.

COPY-READY PROMPT

Use WeeklyEvents, DefectCounts and Evidence. Draft a DMAIC table in Excel: phase, claim, metric, source, limitation, next action. Define the CTQ as delivery-test acceptance deviation, not chemical compliance. Calculate baseline and pilot rates using summed deviations divided by summed opportunities, not a simple average when denominators differ.

Input

Select the named source table

Review

Check the changed cells, formulas and claims

DMAIC2 · Analyse

Copy from the workbook Prompts sheet into Copilot in Excel.

COPY-READY PROMPT

Create a Pareto table and chart from baseline DefectCounts, including cumulative percentage. Reconcile the total with baseline WeeklyEvents. Compare the phase rates and express both percentage-point and relative change. Explain that fewer deviations after two concurrent changes do not isolate the cause or establish sustained improvement.

Input

Select the named source table

Review

Check the changed cells, formulas and claims

DMAIC3 · Improve and control

Copy from the workbook Prompts sheet into Copilot in Excel.

COPY-READY PROMPT

Draft a controlled follow-up trial and ControlPlan table: metric, source, frequency, owner, signal rule, review response and evidence. Link to SPC signals and FMEA actions. Preserve the open removed-strainer review. Do not claim causal proof, savings or operational approval not supported by the source. End with a six-cell management summary and unresolved questions.

Input

Select the named source table

Review

Check the changed cells, formulas and claims

Define the CTQ precisely

Count delivery-test deviations per opportunity.

01

Outcome

Acceptance deviation

02

Denominator

Logged test opportunities

03

Boundary

Not water-quality compliance

Use rates when periods differ

The eight-week log is separate from the120-test investigation.

Baseline

93 /1008 =9.22619%

Pilot

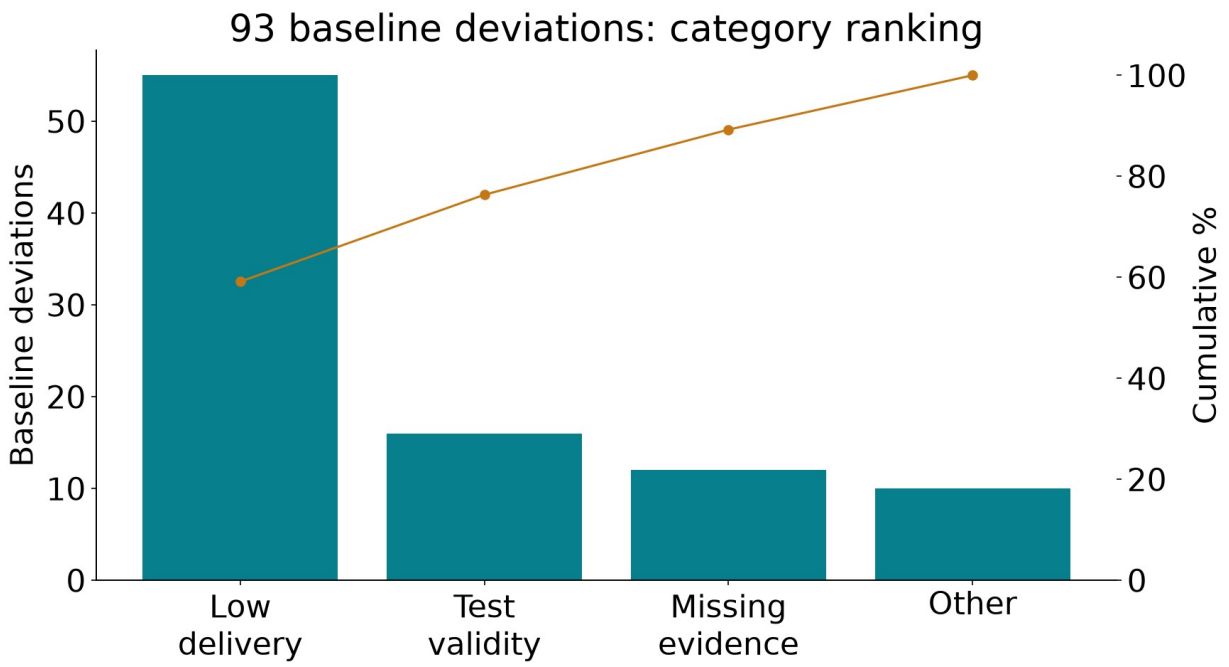
11 /336 =3.27381%

Scope

Six baseline weeks; two pilot weeks

Pareto directs attention

Category counts prioritise investigation; they do not prove causes.



Total

93 baseline deviations

Top category

Low delivery 55

Top two

71 /93 =76.34%

Two ways to express improvement

Label the calculation so percentages are not confused.

Absolute

5.95238 percentage points

Relative

64.51613% reduction

Limit

Short pilot; concurrent changes

Improve with a testable proposal

A before/after difference does not isolate one cause.



Intervention

Strainer replacement and verification

Confounding

Two changes occurred together

Next study

Comparable conditions and reviewed evidence

Make the control plan actionable

Define ownership and response before a new signal appears.

Metric

Delivery and deviation rate

Source

Standard test and opportunity log

Owner

Named responsible role

Response

Evidence review under approved procedures

One workbook supports the review

Every management claim should trace back to a row.

Problem

What is being improved?

Evidence

Baseline, pilot and chart

Limits

What remains unresolved?

Decision

What review is requested?

Compare the prompt with the result

Better prompting is measured at equal quality.

Fidelity

Source facts preserved

Arithmetic

Formulas independently checked

Uncertainty

No invented cause or approval

Effort

Draft plus review and correction

Final workbook review

Show one result and one important limitation.



SPC

Signal and fixed baseline

FMEA

Priority and severity gate

DMAIC

Rate change and next evidence